

Andrey Zaytsev

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Senior Product Security / Application Security Engineer with 10+ years in software and security engineering, including 5.5 years securing Chrome at Google. Expertise in threat modeling, security architecture and design review, secure code review, web/API security, vulnerability remediation, secure SDLC/DevSecOps, and authentication and authorization.

Experience

Independent Product Security Engineer

Jul 2025 - Present

- Designed TutorOps, a security-first Java 21/Spring Boot 4.1 platform with an authorization matrix, security invariants, and deny-by-default role-based access control (RBAC)
- Implemented OAuth 2.0/OpenID Connect Authorization Code with PKCE using Keycloak and Spring Security; validated JWT signatures, issuer, audience, expiry, and role claims
- Built GitHub Actions DevSecOps gates for Maven tests, CodeQL SAST, dependency review/Dependabot, Gitleaks secrets scanning, Trivy SCA, and CycloneDX SBOM generation
- Built and operate solange.top, a production React/TypeScript product on Vercel with Stripe and Cal.com integrations; own application security, deployment, and operations

Chrome Privacy & Security Engineer

Google · Munich, Germany

Jan 2020 - Jun 2025

- Led the end-to-end remediation of a Chrome Android update-detection vulnerability that could suppress warnings for outdated, vulnerable browser versions; redesigned the client/server flow, fixed protocol and threading defects, and drove backend changes across Chrome Update infrastructure teams
- Owned Chrome Android Safety Check from technical design through launch as a prominent top-level Settings entry; obtained security, privacy, and accessibility approvals and kept leadership aligned while unifying compromised-password, Safe Browsing, and browser-update risks in one remediation workflow
- Owned 200+ end-to-end security/privacy launch reviews using data-flow tracing, trust-boundary/abuse-case analysis, architecture review, and threat modeling; drove revisions and escalated 3+ launch blocks, partnering with director-level leadership on mitigations or policy exceptions
- Selected as a Chrome telemetry approver, especially for Android; embedded reusable guidelines in the standard review workflow for 10+ teams, reducing repeat policy violations involving PII or linkable data and accelerating reviews
- Served as Chromium code owner and mobile-development point of contact, conducting 500+ C++/Java/JNI code reviews and surfacing pre-merge security and reliability risks in lifecycle handling, feature gating, API boundaries, and test coverage; mentored an engineer to independent ownership
- Analyzed Chrome Android and CI logs to debug cross-layer failures; in recurring on-call, restored builders, rolled back breaking changes, and helped owners fix and safely reland 20+ changes

Search Infrastructure & Reliability Engineer

Google · Chicago, IL, USA

Oct 2018 - Jan 2020

- Designed and operated Borg-based infrastructure (Google's Kubernetes precursor) and Monarch monitoring across 1,500+ service instances; defined SLAs and alerts and improved release-signal reliability by eliminating configuration drift and flaky infrastructure failures

Software Engineer

Jump Trading · Chicago, IL, USA

Jun 2016 - Jul 2018

- Built large-scale C++/CUDA and Splunk systems, including a 100M-event/day analytics pipeline that cut anomaly diagnosis from hours to minutes and a 500K-write/sec storage engine with sub-millisecond p99 latency

Software Engineer Intern

Dropbox · San Francisco, CA, USA

May 2015 - Aug 2015

- Built and shipped a Python/React/TypeScript distributed system for team-administration actions at 100,000-user scale

Software Engineer Intern

Apple · Cupertino, CA, USA

May 2014 - Aug 2014

- Improved iOS and OS X input-method usability and accessibility in Objective-C using Xcode and platform SDKs

Skills

Application & Product Security: Threat modeling, security architecture/design review, secure code review, secure SDLC, vulnerability remediation, web/API security, OWASP Top 10, Burp Suite, authentication/authorization, OAuth 2.0/OIDC, JWT, RBAC, privacy/data protection

DevSecOps & Engineering: CI/CD, GitHub Actions, SAST, SCA, secrets scanning, dependency review, SBOM; CodeQL, Gitleaks, Trivy, CycloneDX; Java/Spring Boot, Spring Security, Keycloak, C++, Python, TypeScript

AI-assisted Engineering: Agentic development / automated PR review with OpenAI Codex, GitHub Copilot, and CodeRabbit

Languages: English (fluent), Russian (fluent), Spanish (fluent), Portuguese (fluent), German (conversational)

Education

University of Illinois at Urbana-Champaign

Champaign, IL, USA

GPA: 4.0/4.0

Master of Science in Computer Science – Machine Learning

Jan 2017 - Dec 2017

Bachelor of Science in Computer Science

Aug 2013 - Dec 2016

Bronze Tablet Honoree, James Scholar, Dean's List